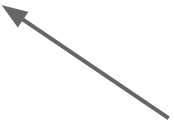


CS-141

clicker questions on functions #1

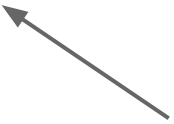
```
public static _____ max(int a, int b) {  
    if (a > b) {  
        return a;  
    } else {  
        return b;  
    }  
}
```



What goes here?

- A. String
- B. void
- C. int
- D. double
- E. none of the above

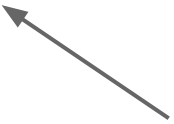
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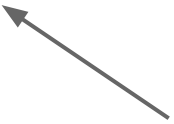
```
public static _____ doubleString(String s)
{
    return s + s;
}
```



What goes here?

- A. String
- B. void
- C. int
- D. double
- E. none of the above

```
public static _____ doubleString(String s)
{
    return s + s;
}
```



What goes here?

- A. String**
- B. void
- C. int
- D. double
- E. none of the above

Do max1 and max2 do the same thing?

```
public static int max1(int a, int b) {  
    if (a > b) {  
        return a;  
    } else {  
        return b;  
    }  
}
```

```
public static int max2(int a, int b) {  
    if (a > b) {  
        return a;  
    }  
    return b;  
}
```

A: Yes!
not A: No!

Do max1 and max2 do the same thing?

```
public static int max1(int a, int b) {  
    if (a > b) {  
        return a;  
    } else {  
        return b;  
    }  
}
```

```
public static int max2(int a, int b) {  
    if (a > b) {  
        return a;  
    }  
    return b;  
}
```

A: Yes!

not A: No!

Does this work?

```
public static int max1(int a, int b) {  
    if (a > b) {  
        return a;  
    } else if (a <= b) {  
        return b;  
    }  
}
```

A: yes!
not A: no!

Does this work?

```
public static int max1(int a, int b) {  
    if (a > b) {  
        return a;  
    } else if (a <= b) {  
        return b;  
    }  
}
```

A: yes!

not A: no!

This fixed it!

```
public static int max1(int a, int b) {  
    if (a > b) {  
        return a;  
    } else {  
        return b;  
    }  
}
```

What does this print?

```
public static int doubleIt(int x) {  
    return x * 2;  
}
```

```
public static void main(String[] args) {  
    int y = doubleIt(3);  
    System.out.println(y);  
}
```

A: y

B: 3

C: 6

D: doesn't compile

E: runs but generates exception

What does this print?

```
public static int doubleIt(int x) {  
    return x * 2;  
}
```

```
public static void main(String[] args) {  
    int y = doubleIt(3);  
    System.out.println(y);  
}
```

A: y

B: 3

C: 6

D: doesn't compile

E: runs but generates exception

Draw this function (static method) on the board:

```
public static void drawRow(IDrawGrid grid, int row,
Color color)
{
    for (int i=0; i<grid.getNumCols(); i++) {
        grid.setColor(row, i, color);
    }
}
```

What draws Indonesia's flag?



- A:** for (int i=0; i<grid.getNumRows()/2; i++) {
 drawRow(grid, i, Color.RED);
 }
- B:** for (int i=0; i<grid.getNumCols()/2; i++) {
 drawRow(grid, i, Color.RED);
 }
- C:** for (int i=0; i<grid.getNumCols()/2; i++) {
 for (int j=0; j<grid.getNumRows(); j++) {
 drawRow(grid, i, Color.RED);
 drawRow(grid, j, Color.RED);
 }
 }
- D:** all of the above
- E:** none of the above

What draws Indonesia's flag?



```
for (int i=0; i<grid.getNumRows()/2; i++) {  
A:   drawRow(grid, i, Color.RED);  
}
```

```
for (int i=0; i<grid.getNumCols()/2; i++) {  
B:   drawRow(grid, i, Color.RED);  
}
```

```
for (int i=0; i<grid.getNumCols()/2; i++) {  
C:   for (int j=0; j<grid.getNumRows(); j++) {  
       drawRow(grid, i, Color.RED);  
       drawRow(grid, j, Color.RED);  
   }  
}
```

D: all of the above

E: none of the above

Write a function (static method) called `drawCol` that takes a `SimpleGrid`, an `int` called `col`, and a `color` as parameters, and sets all of the rows of the given column to the given color.

Basically, it is the opposite of the `drawRow` function we used for the Indonesian flag.

Write a function (static method) called drawCol that takes a SimpleGrid, an int called col, and a color as parameters, and sets all of the rows of the given column to the given color.

```
public static void drawCol(SimpleGrid grid, int col, Color color)
{
    for (int i = 0; i < grid.getNumRows(); i++) {
        grid.setColor(i, col, color);
    }
}
```

Demo Codingbat problems with strings and loops

<https://codingbat.com/prob/p171896>

<https://codingbat.com/prob/p115863>

<https://codingbat.com/prob/p123614>

etc